Progress Check Presentation

Natural Disaster Severity Prediction

Archived draft: this longer deck predates the CatBoost ensemble update. Use `progress-check-2026-05-21-5min.marp.md` or `progress-check-2026-05-21-5min.marp.zh.md` for the current implementation narrative.

Group 5

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Task Description & Dataset Overview

Objective: Predict the weekly drought severity level for 2,248 distinct regions over a 5–week forecasting horizon, following a 91–day blind gap.

Property	Value
Total Regions	2,248
Training Records	12,319,040
Weekly Labeled Targets	1,746,696
Blind Test Period	91 days
Meteorological Features	14

Evaluation Metric: Mean Absolute Error (MAE). Lower is better.

Data Observations & Critical Challenges

The primary challenge lies in the discrepancy between the temporal resolution of features and targets.

- **Sparsity:** The target variable (`score`) is recorded only once per week, rendering approximately 85% of target observations as NaN.
- **Temporal Alignment:** Extracting meaningful long–term trends from daily meteorological signals to predict weekly aggregated severity.
- **The Blind Gap:** The 91–day test period completely lacks ground–truth severity scores. Consequently, direct autoregressive feature construction introduces severe Train–Test Discrepancy.

Current Pipeline Architecture

We have established a reproducible **Direct Horizon Forecasting Pipeline**.

1. **Feature Engineering:** Extract temporal patterns via lag windows, rolling statistics, and exponentially weighted moving averages (EWMA).
2. **Domain-Specific Indicators:** Synthesize specialized meteorological indices, including drought index approximations and dewpoint spreads.
3. **Gap-Aware Score Context:** Use historical score features only after the 91-day blind gap to prevent leakage.
4. **Multi-Horizon Modeling:** Train five independent models—one for each forecasting horizon (Week 1 through Week 5)—to prevent recursive error accumulation.
5. **Ensembling & Tracking:** Support LightGBM, XGBoost, and CatBoost weighted ensembling with strict versioned experiment tracking.

Feature Engineering Strategies

The pipeline supports configurable feature profiles to balance predictive capacity and computational constraints:

- **Meteorological Dynamics:** Short-to-long term rolling means and variances.
- **Temporal Encodings:** Cyclical transformations (sine/cosine) of month, week, and day-of-year to capture seasonality.
- **Domain Synthetics:** Extended drought indices and dryness approximations.
- **Gap-Aware Autoregression:** Historical score features strictly computed prior to the 91-day gap to prevent data leakage.
- **Current Optimal Profile:** The `lean` profile yields the best computational efficiency without sacrificing significant representational power.

Direct Multi-Step Forecasting

Independent regressors are allocated for each temporal horizon (`target_week1` to `target_week5`), optimizing specific lag dependencies and mitigating recursive bias.

Validation Strategy

We transitioned from a random holdout split to a strict **Chronological Holdout Split**. This fundamentally eliminates time-leakage and realistically simulates the 91-day blind forecasting scenario.

Local Validation Results

Key experiments establishing our methodology:

Model	Validation	Features	Local MAE
`lgbm_v2` (Direct Horizon)	Chronological	337	0.6942
`xgb_v1` (Direct Horizon)	Chronological	337	0.7150
`catboost_tail2737` (Direct Horizon)	Rolling Origin	372	0.2212
`lgbm_direct` (Pure Weather)	Chronological	318	0.6770
`xgb_direct` (Pure Weather)	Chronological	318	0.7320

Insight: Validation scores from holdout and rolling-origin modes are not directly comparable; Kaggle public/private results remain the external generalization check.

Kaggle Public Leaderboard (Ablation)

Method	MAE	Key Takeaway
Naive Persistence (ffill)	1.0815	Static baseline fails entirely; drought is highly dynamic.
Misaligned Ensemble	0.8851	Mixing leaky persistence with clean models degrades generalization.
Pure Weather (Strategy B)	0.8640	Excellent Local MAE, but poor Leaderboard generalization.
Long-Term Weather (365d)	0.8604	Fails to replace unobserved confounders in historical scores.

Kaggle Leaderboard (Current Legal Best)

Method	MAE	Key Takeaway
LGB/XGB/CatBoost 35/35/30	0.8124 🏆	**Current best legal public score.** CatBoost adds useful diversity.
LGB/XGB 50/50 Anchor	0.8232	Previous legal anchor; still useful for private leaderboard comparison.
Initial Submission (Leak)	0.8094	Accidental hybrid due to feature artifact. **Discarded to maintain strict academic rigor.**

Static Private Leaderboard Check

The 5/15 static private leaderboard changed our risk assessment.

Signal	Observation	Action
Static private rank	**Team 5: Rank 3**	Competitive, but not safely above Baseline 3.
Baseline 3 pressure	**13 entries** needed to cross Public Baseline 3	Spend submissions only on clear hypotheses.
Public/private gap	Public score improved, private rank still risky	Prioritize robust validation and reproducibility.

Interpretation: Optimize private robustness, not only public leaderboard score.

Best Legal Submission Record

Item	Value
Submission file	`submissions/ensemble_20260516_lgb_xgb_cat2737_35_35_30.csv`
Public MAE	**0.8124**
Components	`lgbm_v2` + `xgb_v1` + `catboost_tail2737`
Blend rule	35% LightGBM / 35% XGBoost / 30% CatBoost
Rows	2,248 regions
SHA-256 prefix	`bee6f618828d`

Reproducibility note: The leaky reproduction run is retained only as a diagnostic artifact and is excluded from valid model selection.

The Kaggle Paradox & Concluding Insights

This extensive ablation study provided a profound insight into the dataset dynamics:

We hypothesized that the inherent noise in historical score reconstruction would degrade performance, prompting us to rely exclusively on pure, long-term meteorological signals (up to 365 days). While this pure approach achieved an outstanding Local MAE of `0.488`, it failed to generalize, scoring only `0.8604` on the Kaggle Leaderboard.

The Kaggle Paradox: Historical drought scores, despite their noise and sparsity, encapsulate critical **Unobserved Confounders**—such as local water infrastructure, agricultural habits, and soil retention characteristics—that pure weather data cannot measure. Consequently, the current direct-horizon ensemble keeps 91-day-gapped historical score context and adds CatBoost diversity, reaching

System Engineering & Infrastructure

The codebase has been refactored to support robust, production-level iteration:

- **Modular Core:** Feature engineering, training, and inference logic are strictly decoupled (`src/train.py`, `src/predict.py`, `src/features.py`).
- **Ensemble Tooling:** Automated post-processing scripts (`src/ensemble.py`) facilitate two-model and three-model blending.
- **Experiment Tracking:** Every execution persists configuration metadata, evaluation metrics, and model weights to a versioned `experiments/` directory.
- **Audit Trail:** Current submission lineage is summarized in `docs/status/experiment_summary.md`.

Next Steps & Future Work

Prior to the final submission deadline, our priorities are:

- **Private–Robust Optimization:** Tune blend weights only when supported by leakage–free validation evidence.
- **Validation Upgrade:** Align local validation more tightly with the 91–day blind gap and private leaderboard behavior.
- **Documentation:** Finalize the IEEE standard report (`report/main.tex`), ensuring all findings from the ablation study and the Kaggle Paradox are thoroughly articulated.

Summary

Achievements:

- Eradicated critical time–leakage and train–test discrepancies.
- Developed a robust, modular Direct Horizon Forecasting pipeline.
- Successfully diagnosed the "Kaggle Paradox," proving the necessity of Unobserved Confounders over pure meteorological data.
- Established a reproducible best legal Kaggle submission: **0.8124 public MAE**.
- Confirmed 5/15 static private leaderboard risk: **Team 5 rank 3, below Baseline 3**.

Current Focus: Improve private robustness and keep code/report/submission artifacts fully consistent.

Thank you (∠·ω)⌒★